Autoclave



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Autoclave; Autoclave - separate for media sterilization and waste disposal; Sterilizers, autoclave

What it is used for

An autoclave uses pressurised saturated steam to sterilise suitable laboratory materials or treat designated biological waste. Correct loading, validated cycles, and careful pressure release are essential for effective treatment and operator safety.

Main parts

- Sterilisation chamber holds the materials being treated.
- Door and locking mechanism seal the chamber during operation.
- Steam generator or steam supply provides the required steam.
- Temperature and pressure sensors monitor cycle conditions.
- Control panel selects the cycle and displays operating status.
- Drain and exhaust system remove condensate and release steam through the designed pathway.
- Gaskets and safety interlocks help maintain a secure seal.
- Trays, racks, and containers support the load.

Before you start

1. Only trained, authorised personnel should operate the autoclave under the laboratory's approved procedures.
2. Read the manual and select the validated cycle for the specific load.
3. Inspect the chamber, door gasket, locking mechanism, drain, and indicators for damage or blockage.
4. Check that water supply and other required utilities are available and correct.
5. Use autoclave-compatible containers and arrange loads to allow steam circulation.
6. Confirm biological waste is packaged and segregated according to laboratory procedures and applicable rules.

How to use

1. Wear the specified PPE, including heat-resistant gloves and eye or face protection as required.
2. Prepare and load materials in compatible containers without overfilling or blocking steam pathways.
3. Place liquid containers in suitable secondary containment and loosen closures as required by the validated procedure.
4. Close and lock the door using the manufacturer's specified method.
5. Select the approved cycle for the load and verify the displayed settings.

6. Start the cycle and monitor temperature, pressure, and cycle status without opening the chamber.
7. Wait until the cycle completes and pressure returns to the permitted safe level.
8. Open the door cautiously from the side, allowing residual steam to escape away from the face.
9. Remove the load with suitable heat protection and allow items to cool in a designated area.
10. Document the cycle, indicators, and any required chemical or biological monitoring results; report failures before using the load.

⚠ Safety

- Never open the door while the chamber is pressurised or the cycle is active.
- Do not bypass door locks, pressure controls, or safety interlocks.
- Use suitable heat-resistant PPE and keep exposed skin away from steam and hot surfaces.
- Do not autoclave sealed containers, volatile chemicals, flammable materials, or incompatible substances.
- Treat liquid loads carefully because hot liquids may boil over after the cycle.
- If the cycle fails or an alarm occurs, isolate the unit as instructed and notify authorised personnel.

Care & storage

- Clean the chamber and trays after cooling, following the manufacturer's procedure.
- Inspect the gasket, drain filter, and door mechanism at recommended intervals.
- Keep records of maintenance, cycle verification, and required performance testing.
- Arrange authorised servicing for leaks, unusual noises, failed indicators, or repeated cycle errors.

Common mistakes to avoid

- Selecting a cycle that is unsuitable for the load or container type.
- Overloading the chamber or blocking steam circulation.
- Opening the door before pressure has returned to a safe level.
- Using treated materials without checking required cycle records or monitoring results.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.